Advancements and applications of position-sensitive detector (PSD): a review

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(Received 1 July 2023; Revised 17 October 2023)

This paper presents a review of the position-sensitive detector (PSD) sensor, covering different types of PSD and recent works related to this field. Furthermore, it explains the theoretical concepts and provides information about its structure and principles of operation. Moreover, it includes the main information about the available commercial PSDs from different companies, along with a comparison between the common modules. The PSD features include high position resolution, fast response, and a wide dynamic range. These features make it suitable for various fields and applications, such as imaging, spectrometry, spectroscopy and others.

Document code: A **Article ID:** 1673-1905(2024)06-0330-9

DOI https://doi.org/10.1007/s11801-024-3117-2

Visible light communication (VLC) is a new wireless communication technique that transmits data using visible light as a medium[1,2]. It leverages the properties of light, typically from light-emitting diodes (LEDs), to achieve high-speed communication between devices in both directions[3,4]. It has various features, such as a high data rate, security, immunity to radio interference, and energy efficiency[5,6].

The main problem facing the VLC is alignment[7]. To establish a successful VLC system, both the transmitter and the receiver should be in the same line-of-sight (aligned)[8]. Otherwise, the entire communication system might collapse[9]. Shaking can happen because of wind or maybe the movement of water in the case of underwater wireless optical communication (UWOC)[10]. To overcome this problem, tracking systems can be used, which helps in achieving good signal transmission[11].

Tracking systems play a significant role in VLC by enabling efficient and reliable data transmission between the light source (transmitter) and the receiving device[12]. These systems help maintain a stable connection and optimize the performance of VLC.

A tracking system in a VLC consists of sensors, detectors, or receivers that can detect and track the position or direction of the VLC transmitter or light source. These systems utilize various elements, such as photodiodes, image sensors, or cameras. However, the position-sensitive device (PSD) is the most common element used for tracking in VLC systems[13].

In addition to tracking and alignment systems, PSDs are also used in applications where accurate position sensing is required[14], such as robotic vision systems, medical imaging, building deformation[15] and scientific instrumentation[16]. They offer features like high position resolution, fast response, a wide dynamic range, the ability to measure position in real time, compactness and lightweightness. However, PSDs have some limitations. These limitations mainly include environmental interference, where external factors can affect the accuracy and reliability of the position sensing process; non-linear response characteristics[17]; and temperature and environmental factors that affect the performance of PSDs. Moreover, some PSDs have a limited active area, which may restrict the size of the incident radiation that can be detected.

This review paper delves into the intricacies of PSD technology, exploring its structure and principles of operation, measurements, and resolution enhancement techniques. We present the evolution of PSD-based tracking systems, showcasing their efficacy in diverse scenarios. Furthermore, we talk about employing genetic algorithms (GAs) with PSDs, harnessing the power of computational intelligence to enhance their capabilities. In the era of cutting-edge materials, we unveil the potential of two-dimensional (2D) materials-based PSDs, paving the way for novel sensing paradigms. Additionally, we delve into the concept of reconfigurable multi-functional PSDs, showcasing their adaptability to varied tasks. Finally, we provide insights into the realm of commercial PSDs with extensive comparisons to the available modules. The organization and division of the paper are illustrated in Fig.1.

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A PSD typically consists of a photosensitive area divided into multiple segments or pixels, with some electrodes arranged in a specific pattern[18].

The principle of operation of a PSD is based on the detection of incident light or radiation and the conversion of its position into an electrical signal. When light or radiation strikes the surface of the PSD, it generates charge carriers (electrons or holes) in the material. The distribution of these charge carriers across the electrodes or segments is based on the place of the radiation spot[18].

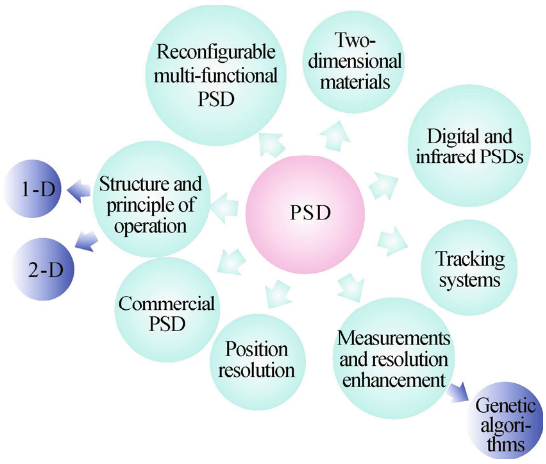

Fig.1 Organizing and division of the paper

By measuring the amount of charge or current generated at each electrode, the PSD can determine the position of the incident light spot in one or two dimensions. This can be achieved using various techniques, such as charge division, resistive readout, or capacitive readout. In short, the exact location of the incident light on the photosensitive surface can be determined by measuring the relative signal strengths or voltage outputs from the electrodes[19].

One-dimensional (1D) PSD has a long, narrow photosensitive area and measures the position of incident light or radiation along a single axis. It offers some features, such as a fast response time and high resolution. Fig.2 illustrates the photosensitive area of a 1D PSD. The position conversion formula can be calculated from Eq.(1) (for example, point A)[18].

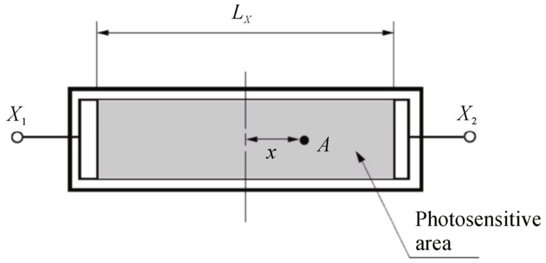

Fig.2 Photosensitive area of 1D PSD

$$X_A = \frac{I_{X_2} - I_{X_1}}{I_{X_1} + I_{X_2}} \times \frac{L_X}{2}, \tag{1}$$

where X_A is the distance from the electrical center of PSD to the light incident position, I_{X_1} is the output current from electrode X_1, I_{X_2} is the output current from electrode X_2, and L_X is the length of the active area.

The 1D PSD is commonly used in many applications, such as linear displacement measurement, alignment systems, and motion control. For example, in optical encoders, 1D PSDs detect the location of a moving object or track the position of a linear stage. Moreover, 1D PSD can be used to measure the readout of an interferogram, where interferometry is a technique that utilizes interference patterns to extract information about the characteristics of light waves or other types of waves.

The photosensitive area of a 2D PSD is illustrated in Fig.3. A 2D PSD practically behaves like a 2D array of photodiodes. So, there are two axes of coordinates, X and Y. The dimension between the two electrodes is L, and the output photocurrents of the electrodes X_1, X_2, Y_1, and Y_2 are I_{X_1}, I_{X_2}, I_{Y_1}, and I_{Y_2}, respectively. The position conversion formula can be calculated from Eq.(2) and Eq.(3) (for example, point A)[20].

$$X_A = \frac{(I_{X_2} + I_{Y_1}) - (I_{X_1} + I_{Y_2})}{I_{X_1} + I_{X_2} + I_{Y_1} + I_{Y_2}} \times \frac{L_X}{2}, \tag{2}$$

$$Y_A = \frac{(I_{X_2} + I_{Y_2}) - (I_{X_1} + I_{Y_1})}{I_{X_1} + I_{X_2} + I_{Y_1} + I_{Y_2}} \times \frac{L_Y}{2}, \tag{3}$$

where Y_A is the distance from the electrical center of PSD to the light incident position according to the Y axis, and L_Y is the length of the active area according to the Y axis.

The PSD (2D) outputs are four signals (currents), their values vary depending on the location of the incident light spot. After the processing of the signals, the accurate position of this beam can be determined, and then suitable action can be taken[21].

In Ref.[18], the authors effectively used MATLAB for modeling and simulating PSD sensors in both 1D and 2D configurations. The models included the simulation of a 1×3 array of 1D PSDs and a 3×3 array of 2D PSDs. Moreover, they provided a comprehensive explanation of the PSD sensor's structure and operational principles.

The resolution of the PSD can be defined as the least displacement of a light spot that can be detected[22]. The minimum detectable position displacement Δx is given by

$$\Delta x = L \times \frac{I_{\mathrm{n}}}{I_{\mathrm{tot}}}, \tag{4}$$

where L represents the photodetector length, I_{tot} expresses the total photocurrent obtained from all electrodes, and I_{n} represents the current generated by the noise.

I_{n} is given by[23]

$$I_{\mathrm{n}} = \sqrt{I_{\mathrm{s}}^2 + I_{\mathrm{T}}^2 + I_{\mathrm{en}}^2}, \tag{5}$$

where I_{en} expresses the noise current that is produced by the amplifiers, I_{s} is the shot noise current, and I_{T} represents the thermal noise current. I_{en} can be calculated from

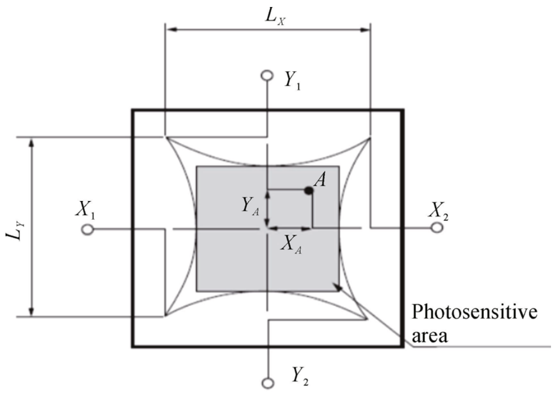

Fig.3 Photosensitive area of 2D PSD

$$I_{en}=\frac{V_{en}}{R_{ie}}\times\sqrt{B}, \tag{6}$$

where V_{en} represents the equivalent noise input voltage of the amplifier (V/Hz$^{1/2}$), R_{ie} expresses the internal electrode resistance of the PSD, and B expresses the bandwidth. I_s can be calculated from[23]

$$\left\langle i_s^2\right\rangle=2qB(I_p+I_B+I_L), \tag{7}$$

where I_p represents the generated photocurrent, I_B is the current generated by the background radiations, I_L is the leakage current, and q is the electron charge. I_T can be calculated from

$$\left\langle i_T^2\right\rangle=4kTB(\frac{1}{R_{eq}}), \tag{8}$$

where k is the Boltzmann's constant, T represents the temperature measured in Kelvin, and R_{eq} represents the equivalent resistance calculated from

$$\frac{1}{R_{eq}}=\frac{1}{R_{sh}}+\frac{1}{R_L}+\frac{1}{R_i}, \tag{9}$$

where R_{sh} is the diode junction resistance, R_L represents the external load resistance, and R_i is the input resistance of the amplifier.

Most research related to the PSD focuses on the measurements and how to enhance the resolution. KIM et al[24] used two PSDs to develop a gap measurement system for plasma display panels (PDPs). One PSD is tasked with identifying the location of the reference beam, which bounces off the upper surface of the gap. Meanwhile, the other PSD is responsible for pinpointing the position of the signal beam, which reflects off the lower surface of the gap. The resolution of the gap dimension was 0.5 μm, and the detecting range was 500 μm. SOLAL et al[25] presented modeling and simulation of PSD using CAO AWB/Cadence. They predicted a specific estimate of both the linearity and spatial resolution of the PSD with its associated electronics.

WANG et al[26] presented the modeling and nonlinear correction approach for the 2D photoelectric PSD using a radial basis function (RBF) neural network. The proposed modeling achieved high linearity and better measurements of the PSD. SOLAL[27] demonstrated a simulation model of duo-lateral PSD to study and analyze its behavior, especially the distortions. He used this model to optimize the sheet resistance of the PSD used for the gamma camera to detect body tumors.

RAHIMI et al[28] proposed improving PSD measurement accuracy using diverse correcting techniques and filters. They used the signal averaging technique to enhance the quality of the signal to noise. Moreover, RAHIMI et al developed different methods for determining laser blocking positions. The most precise technique involves monitoring the minimal alterations in two photo-current electrodes situated on opposite sides of the PSD, which is known as the inverse correlation. In short, when one of them increases, the other one decreases.

NIU et al[29] performed the simulation for a novel density well-logging tool using 1D PSD by the Monte Carlo approach. However, this simulation ignored some factors like the nonlinearity of the NaI crystal, the circuit noise, and the temperature effect.

ZHANG et al[16] developed a method to enhance the accuracy of lasers using 2D PSD. The approach is very easy and flexible, with good accuracy for the nonlinear correction of the PSD. To improve the measurement data, they use the error-curved surface interpolation approach, which reduces the entire position error by about 32%.

Recently, the GA was used in much research related to PSD-based positioning systems[14] because it is considered as a powerful approach to optimize and enhance the accuracy of these systems[30]. It can optimize the calibration parameters of the PSD sensor array, improving the accuracy of position measurements. GA can also develop error correction models that compensate for environmental conditions, sensor imperfections, or noise in the PSD data, which reduces positioning errors and enhances the overall system reliability. Furthermore, it can be used to optimize the placement and orientation of PSD sensors within a system. This optimization ensures that the sensors are strategically positioned to maximize accuracy and coverage in the measurement area.

LLANA-CALVO et al[14] proposed a high-precision indoor positioning system depending on LED illumination and a PSD sensor. They used the GA to obtain the model parameters of the PSD. In Ref.[31], the authors also employed a GA to conduct a basic and weak calibration method for an in-plane switching system that depends on a PSD. Their comparative error analysis revealed that the most substantial decrease in accuracy, when compared to the considerably more expensive formal calibration approach, was less than 25 mm. Furthermore, they determined that the overall absolute positioning error remained below 35 mm, while orientation errors were approximately 0.25°. QU et al[32] used the pin-cushion PSD in the solar occultation flux (SOF) method, which is based on the Fourier transform infrared spectrometer (SOF-FTIR) system for solar tracking closed-loop control. They analyzed and studied the positioning error factors of the PSD as well as the noise factors.

In the digital PSD field, MASSARI et al[33] presented a fast digital 2D PSD, which is an array with 20×20 pixels and a pitch of 70 μm. The fill factor is 15%, and the die area is about 8 mm^2 including the pads. It is designed using 0.8 μm CMOS technology. The maximum power consumption is about 5 mW.

MAKYNEN et al[34] proposed a digital PSD. They implemented three digital PSDs. The first one was a 16×16-pixel array and was implemented using 1.2 μm CMOS technology. The second one was a 32×32-pixel array and manufactured using 0.8 μm CMOS technology. The third one was a 128×128-pixel array and designed using 0.6 μm CMOS technology.

An infrared PSD (IRPSD) is a specialized type of PSD. It operates in the infrared portion of the electromagnetic spectrum, typically covering wavelengths longer than those of visible light. IRPSDs are used in a wide range of applications, including thermal imaging, and detecting and counting objects[35].

TAKAHATA et al[36] proposed a low-cost and high-performance IRPSD, which is an infrared array sensor. Here, the PSD is used to detect the position of objects with different temperatures as well as calculate the total number of objects without the need for digital image processing (DIP). The proposed sensor has a better signal-to-noise ratio than the regular infrared array sensors as well as the single-element infrared sensors.

An enhanced IRPSD designed to function within a vacuum environment is presented in Ref.[37]. This IRPSD is a 2D infrared array sensor featuring two thermometers within each pixel. One thermometer is connected in series along rows, while the other is connected in series along columns. This serial connection of thermometers allows for the summation of signals across all pixels within each row and column. This unique array structure enables the IRPSD to determine the position of an object and quantify the quantity of objects present without the need for digital image processing. The fabricated array was 6×6-pixel. They attained a responsivity of 170 V/W in the wavelength range from 8 μm to 14 μm.

Mechanical tracking systems to maintain optical alignment have a deep history and have been widely used in many applications[38,39]. There are many types of these systems, but systems depending on the gimbal are very common. There will be a rotating gimbal where the sensor or PSD is mounted, and then this gimbal moves in one axis, two axes and three axes[40]. HARRIS et al[41] demonstrated the alignment of a communication link utilizing free-space optical technology, which connects a ground station with an unmanned aerial vehicle (UAV). XU et al[42] used PSD to design a tracking system to detect and track sunlight accurately. In other research, proportional, integral, and differential (PID) controlling methods were used to achieve laser tracking systems[43]. HEWEAGE et al[20] also used the PSD (quadrant detector) to perform a laser beam spot position determination (LBSPD) circuit. A quadrant detector typically consists of four separate photodiodes or light-sensitive elements arranged in a quadrant. It does not give an accurate position of the incident light; it only gives in which quarter it is. The authors in Ref.[20] used the ORCAD-PSpise software tool to achieve the modeling and simulation.

MALLICK et al[44] employed the PSD to develop an effective tracking technique for an optical communication system. In this system, the free-space optical (FSO) link spanned a distance of more than 500 m, and the baseband transmission rate reached approximately 10 Gbit/s for over a 500-m-long FSO link.

DWIK et al[45] used the PSD to design a simple laser tracking system where the signal conditioning circuit was implemented using a programmable system on chip (PSoC) to be used in VLC. The PSoC is an advanced microcontroller that integrates configurable analog and digital circuits.

The common materials used to manufacture the PSDs are silicon, gallium arsenide, and indium gallium arsenide. Besides, perovskite[46] has gained significant attention in recent years for its excellent optoelectronic properties. However, novel PSDs have appeared recently that are manufactured using 2D materials. These photodetectors utilize the unique properties of 2D materials to achieve low-power consumption, high spatial resolution and high sensitivity[47]. Many materials are used, such as graphene and transition metal dichalcogenides (TMDs), like molybdenum disulfide (MoS_2). Tab.1 shows a comparison between 2D material-based PSDs.

Tab.1 Comparison between 2D materials-based PSDs

Ref.	Materials	Spectral range (nm)	Operating area (mm)	Spatial sensitivity (mV/mm)	Response time (μs)
[48]	$G/SiO_2/Si$	514	4	-	1.2
[49]	G/Si	532—1 550	8	43	0.44
[50]	G/Si	633—750	5	60	10
[51]	G/Ge	750—1 600	5	47	2.6
[52]	MoS_2/Si	405—980	1.6	183	2.1
[53]	$MoS_2/SiO_2/Si$	300—1 000	1	355.4	16.2
[54]	$MoSe_2/Si$	405—980	1.6	563	2
[55]	SnSe/Si	532—808	1.2	250	2
[56]	$MoS_2/GaAs$	650	1	416.4	-

DWIK et al[57] proposed a configurable multi-functional PSD that can achieve positioning, energy harvesting and optical communication. The authors proposed to modify the architecture by using transistors to switch between the two modes of operation of the photodiodes (photoconductive and photovoltaic). Furthermore, switching between the output pins could be achieved depending on the desired function. The proposed sensor can extend the battery's lifetime, increase the integration and functionality of the system, and reduce its physical size. Simulation using MATLAB was performed to examine the functionality of the sensor, and the results were compatible with the hypothesis. It is expected that this type of sensor will make a quantum leap in optical sensors and VLC systems. Fig.4 depicts the proposed architecture of the sensor[57].

Commercially, PSDs are readily available sensors that can be purchased from various manufacturers and suppliers. There is a wide range of modules with different specifications, suitable for a wide range of applications. The prices can range from a few hundred dollars to several thousand dollars. There are many companies that produce this type of sensor, such as Hamamatsu. The company was founded in 1953 and has since become a leading global provider of photodetectors, imaging devices, light sources, and related products. Thorlabs is an American company that was founded in 1989. It is known for its extensive range of products related to optics and photonics. Their product offerings include lasers, optics, optomechanical components, photodetectors, optical fibers, imaging systems, and many other items used in research and industrial applications. OSI Optoelectronics is also an American company that was founded in 1967. It offers a wide range of optoelectronic products, including photodiodes, photodetectors, arrays, and specialized sensors. Tab.2 shows some commercial PSD modules specifications that are manufactured by different companies. Fig.5 shows some commercial PSD sensors.

In short, PSD is a common optical sensor which is widely used in various applications that require precise position measurement. However, like any technology or sensor, PSDs have advantages and disadvantages. Tab.3 summarizes the main features and drawbacks of the PSD.

The main advantages of PSDs include the following. First, they have a fast response time[58,59], allowing them to capture and track rapid changes in position. Second, PSDs typically have a relatively large active area compared to other position sensing technologies[46]. Third, PSDs can operate over a wide spectral range[60], including ultraviolet, visible, and infrared wavelengths. This flexibility makes them compatible with various light sources and enables their use in diverse applications across different industries. In addition, PSDs are non-contact devices, meaning they do not require physical contact with the object being measured. This non-intrusive nature makes them suitable for applications where contact could be detrimental, such as delicate surfaces, moving objects, or situations where cleanliness is important. Furthermore, PSDs generally consume low power, which makes them appropriate for low-power applications.

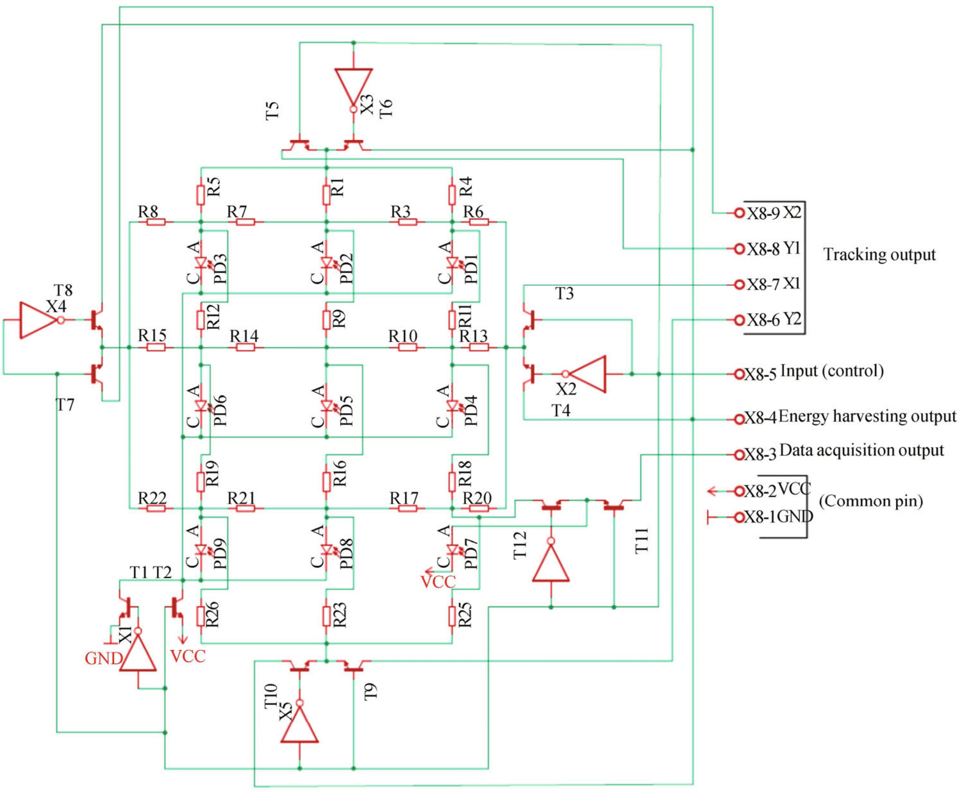

Fig.4 Schematic diagram of the proposed PSD in Ref.[57]

Tab.2 Comparison between some commercial PSDs modules manufactured by different companies

Module	Photosensitive area (mm)	Dark current (nA)	Interelectrode resistance (kΩ)	Spectral range (nm)	Rise time (ns)	Package material	Type	Company
S4583-04	1×3	1	140	760—1 060	10 000	Plastic	1D	Hamamatsu
S4584-06	1×3.5	1	140	320—1 100	3 000	Plastic	1D	Hamamatsu
S8543	0.7×24	15	140	320—1 100	20 000	Ceramic	1D	Hamamatsu
S3274-05	1×3.5	1	400	760—1 100	15 000	Plastic	1D	Hamamatsu
S7105-04	1×4.2	2	140	760—1 100	5 000	Plastic	1D	Hamamatsu
S7105-06	1×4.2	2	140	320—1 100	5 000	Plastic	1D	Hamamatsu
S3931	1×6	10	50	320—1 100	1 500	Ceramic	1D	Hamamatsu
S3932	1×12	20	50	320—1 100	3 000	Ceramic	1D	Hamamatsu
S8543	0.7×24	15	140	320—1 100	20 000	Ceramic	1D	Hamamatsu
S1880	12×12	1	10	320—1 060	1 500	Ceramic	2D	Hamamatsu
S2044	4.7×4.7	0.5	10	320—1 060	300	Metal	2D	Hamamatsu
S5990-01	4×4	10	7	320—1 100	1 000	Ceramic	2D	Hamamatsu
S5991-01	9×9	50	7	320—1 100	2 000	Ceramic	2D	Hamamatsu
PDA20C2	3.14×3.14	-	-	800—1 700	70	Plastic	2D	Thorlabs
PDA10D2	0.8×0.8	-	-	900—2 600	15	Plastic	2D	Thorlabs
PDA05CF2	0.2×0.2	-	-	800—1 700	2.3	Plastic	2D	Thorlabs
SL3-1	1×3	5	15	400—1 100	40	Metal	1D	OSI Optoelectronics
SL5-2	1×5	10	20	400—1 100	100	Ceramic	1D	OSI Optoelectronics
SL15	1×15	150	60	400—1 100	600	Ceramic	1D	OSI Optoelectronics
DL-2	2×2	30	5	400—1 100	30	Metal	2D	OSI Optoelectronics
DL-10	10×10	500	5	400—1 100	200	Metal	2D	OSI Optoelectronics
DLS-20	20×20	100	5	400—1 100	1 200	Ceramic	2D	OSI Optoelectronics
DL-20C	20×20	2 000	5	400—1 100	1 000	Ceramic	2D	OSI Optoelectronics

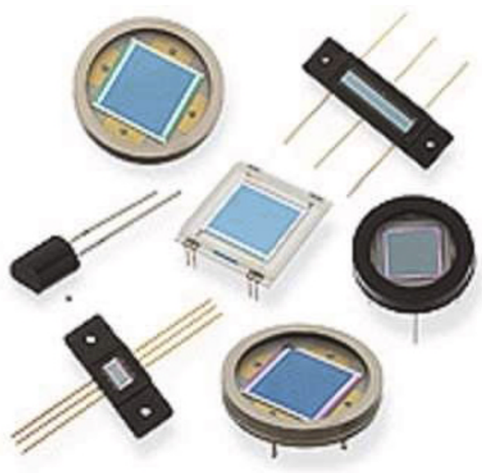

Fig.5 Some commercial PSD sensors modules

Tab.3 Advantages and disadvantages of the PSD

Advantages	Disadvantages
Fast response time	Nonlinearity
Large active area	Environmental interference
Wide spectral response	Complex circuitry
Contactless and non-intrusive	Cost
Low power consumption	Non-uniformity and dead areas

On the other side, PSDs have many drawbacks, which can be listed as follows. First, PSDs may exhibit nonlinearity in their response. This means that the output current of the photodetector may not be in proportion to the position of the incident light spot. Nonlinearity can introduce errors and require additional calibration or compensation techniques to achieve accurate position measurements. Second, PSDs can be susceptible to environmental interference such as ambient light, electromagnetic noise, or temperature variations. These external factors can affect the accuracy and stability of the position measurement. Shielding, filtering, or other techniques may be required to minimize the impact of such interference. Third, PSDs often require complex readout circuitry to convert the position-dependent signal into a useful output format[17]. The circuit includes different components and functionalities such as current-to-voltage conversion, amplifying, position calculation, feedback and control, and many others. Furthermore, PSDs are considered expensive compared to some alternative position sensing technologies. As mentioned before, the cost of a single PSD can reach up to a few thousand dollars. This cost factor may limit their adoption in certain applications or drive the search for more cost-effective alternatives.

Moreover, non-uniformity and dead areas are a serious drawback in the structure of the PSD[61], where variations in the detector's material properties, readout electronics, and environmental factors can cause non-uniformities, leading to position inaccuracies and reduced spatial resolution. Furthermore, PSDs may have inactive areas or dead regions where position information cannot be accurately obtained[62,63]. These dead areas can arise due to electrode gaps, shielding structures, or defects in the detector's construction. Inefficiencies in charge collection or light detection can also affect the detector's overall sensitivity.

Despite these drawbacks, PSDs remain widely used in many fields, such as robotics, industrial automation, laser alignment, and scientific research, due to their unique capabilities for precise position sensing.

A detailed survey of PSD was presented in this paper. It included the main works related to the PSD, along with an explanation of the theoretical concepts and basis. Furthermore, a comparison between the advantages and disadvantages of the PSD was discussed. It is clear that PSD is an important optical sensor that is used in lots of applications and fields. PSDs are expected to be more integrated and have higher resolution, calling for new designs and structures. So, there are many aspects that need to be investigated, like the materials and the architecture, which can enhance the performance and increase the integration. For instance, using 2D materials and designing reconfigurable architecture achieved good results and seem like a nice field to be investigated in the near future.

Ethics declarations

Conflicts of interest

The authors declare no conflict of interest.

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